Energy Resources

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📜📜 History

Human civilisation has been shaped by the energy resources available to each era.

The Pre-Industrial Era (before 1760): Humans relied on **biomass** (wood for heating, food for muscle power), **wind** (sails, windmills), and **water** (water wheels). Energy was limited and local.

The Industrial Revolution (1760–1840): The invention of the steam engine by **James Watt** (1736–1819) enabled widespread use of **coal**. This was the first time humans could access the stored chemical energy of fossilised plants from millions of years ago. This energy surplus drove factories, railways, and steamships.

The Oil Era (1900–present): The discovery and extraction of **petroleum** provided an even more energy-dense fuel. It powered automobiles, aircraft, plastics, and the modern chemical industry.

The Renewable Transition (2000–present): Growing awareness of climate change and finite fossil fuel reserves has driven a global shift toward renewable energy sources.

Did you know? All fossil fuels (coal, oil, natural gas) are the remains of ancient organisms — coal from trees and plants that died 300 million years ago, oil from tiny marine organisms. This is why they're called "fossil" fuels — they are literally buried sunlight from the Carboniferous period.

🌍 🌍 Nature & Applications

- ⛽ **Petrol/Diesel** power most of the world's cars and trucks
- 💡 **Electricity** is generated from a mix of coal, gas, nuclear, hydro, wind, and solar
- ☀️ **Solar power** is the fastest-growing energy source globally
- 🌊 **Tidal and wave power** harness the Moon's gravitational pull
- 🌡️ **Geothermal** uses heat from inside the Earth (radioactive decay in the core)
- 🪵 **Biomass** (wood, crop waste) is still the primary energy source for ~2 billion people

⚛️ ⚛️ Theory

Non-Renewable Energy Resources

These are resources that are **limited** and will eventually run out.

Resource	How it works	Advantages	Disadvantages
Coal	Burned to heat water → steam → turbine → electricity	Cheap, abundant, established infrastructure	CO_2 emissions, acid rain, mining damage
Oil	Refined into petrol/diesel, burned in engines or power plants	High energy density, versatile (plastics, chemicals)	CO_2 emissions, oil spills, finite
Natural gas	Burned for heating, cooking, electricity	Cleaner than coal/oil (less CO_2 per kWh)	Still produces CO_2, leaks methane (worse GHG)
Nuclear	Uranium fission → heat → steam → turbine → electricity	No CO_2 emissions, huge energy from small fuel	Radioactive waste, disaster risk (Chernobyl, Fukushima)

Renewable Energy Resources

These resources are **replenished naturally** and will not run out.

Resource	How it works	Advantages	Disadvantages
Solar	Photovoltaic cells convert sunlight directly to electricity	Free fuel, no emissions, works at small scale	Intermittent (night/clouds), low efficiency (~20%)
Wind	Wind turns blades → generator → electricity	Free fuel, no emissions, cheap in windy locations	Intermittent (wind varies), visual noise, bird strikes
Hydroelectric	Water flows through dam → turbine → electricity	Reliable (can be turned on/off), long lifespan	Habitat destruction, high upfront cost, drought risk
Tidal	Tidal flows turn underwater turbines	Predictable (tides are regular), no fuel cost	Expensive, limited suitable locations, marine ecosystem impact
Wave	Wave motion drives generators	Huge potential, continuous resource	Technology still developing, harsh marine environment
Geothermal	Heat from underground → steam → turbine	Constant (not intermittent), small land footprint	Only works in geologically active areas, can trigger earthquakes
Biomass	Burn organic matter (wood, crops, waste)	Uses waste materials, carbon-neutral if regrown	Still produces CO_2 and particulates, land competition with food

Comparing Energy Resources

Environmental impact:

- Fossil fuels produce CO_2 (greenhouse gas) and other pollutants
- Nuclear produces radioactive waste

- Renewables produce no direct emissions during operation (but have manufacturing impacts)

Reliability:
- Fossil fuels and nuclear are **reliable** (can generate 24/7)
- Solar and wind are **intermittent** (dependent on weather and time of day)
- Hydro, tidal, and geothermal can provide **base load** or be turned on/off as needed

Energy density:
- Nuclear has the highest energy density (1 kg uranium ≈ 20 000 kg coal)
- Fossil fuels have high energy density
- Renewables generally have low energy density (need large areas)

The Energy Mix

Different countries use different mixes of energy resources depending on:
- **Geography** (sunny? windy? mountains for hydro? geothermal?)
- **Economics** (what's cheapest available?)
- **Politics** (energy security, climate targets)
- **Technology** (grid capacity, storage solutions)

Energy Efficiency in Context

The most environmentally friendly energy is the energy we **don't use**:

Energy conservation methods:
- **Insulation** — reduce heating/cooling needs in buildings
- **LED lighting** — uses ~80% less energy than incandescent bulbs
- **Energy-efficient appliances** — A+++ rated refrigerators, washing machines
- **Public transport** — reduces energy per person per km compared to cars

$$\text{Sustainable energy} = \text{available} + \text{affordable} + \text{environmentally acceptable}$$

$$\text{Energy saved} = \text{energy not generated}$$

✍️ ✍️ Worked Examples

Example 1: Comparing CO_2 Emissions

A coal power station produces 1.0 kg of CO_2 per kWh of electricity. A gas power station produces 0.4 kg of CO_2 per kWh.

How much CO_2 does each produce in a day, if they both generate 500 MW (500 000 kW)?

Solution (Coal):

Solution (Gas):

Gas produces less than half the CO_2 of coal for the same electricity output.

Example 2: Solar Panel Efficiency

A solar panel with area 2 m² receives 1000 W/m² of sunlight. It produces 320 W of electrical power. What is its efficiency?

Solution:

Example 3: Wind Farm Output

A wind turbine has a rated power of 2 MW. It operates at 30% capacity factor (due to varying wind). How much energy does it produce in one year?

Solution:

Example 4: Comparison Table

A household uses 4000 kWh of electricity per year. Compare the CO_2 emissions if this comes from coal vs solar (solar has zero operational emissions, but manufacturing emits ~50 g CO_2 per kWh).

Solution (Coal):

Solution (Solar):

Solar produces **20 times less** CO_2 over its lifetime.

$$\text{Energy per day} = 500\,000 \times 24 = 12\,000\,000 \text{ kWh}$$

$$CO_2 = 12\,000\,000 \times 1.0 = 12\,000\,000 \text{ kg} = 12\,000 \text{ tonnes}$$

$$CO_2 = 12\,000\,000 \times 0.4 = 4\,800\,000 \text{ kg} = 4\,800 \text{ tonnes}$$

$$\text{Input power} = 1000 \times 2 = 2000 \text{ W}$$

$$\text{Efficiency} = \frac{320}{2000} \times 100\% = 16\%$$

$$\text{Actual average power} = 2 \times 0.30 = 0.6 \text{ MW} = 600 \text{ kW}$$

$$\text{Hours per year} = 365 \times 24 = 8760 \text{ h}$$

$$\text{Energy} = 600 \times 8760 = 5\,256\,000 \text{ kWh} = 5.26 \text{ GWh}$$

$$CO_2 = 4000 \times 1.0 = 4000 \text{ kg} = 4 \text{ tonnes}$$

$$CO_2 = 4000 \times 0.05 = 200 \text{ kg} = 0.2 \text{ tonnes}$$